Project Title

Customer Shopping Analysis and Sales Insights Dashboard

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1. Project Overview

Objective

The primary objective of this capstone project is to analyze customer shopping behavior using transactional data and derive actionable business insights through **SQL**, **Python (EDA & data cleaning)**, and **Power BI dashboards**.

Business Problem Statement

Retail businesses need to understand: - Customer purchasing patterns - Product category performance - Revenue contribution by demographics - Seasonal and behavioral trends

This project transforms raw customer shopping data into meaningful insights to support data-driven decision-making.

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2. Tools & Technologies Used

Tool	Purpose
SQL	Data querying, joins, aggregations
Python (Pandas, NumPy, Matplotlib)	Data cleaning, EDA, feature analysis
Power BI	Data modeling, KPIs, dashboards
CSV	Raw and cleaned datasets

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3. Data Sources

3.1 Raw Dataset

- **File Name:** customer_shopping.csv
- **Format:** CSV
- **Description:** Raw transactional customer shopping data

3.2 Cleaned Dataset

- **File Name:** customer_shopping_clean.csv
- **Format:** CSV
- **Description:** Preprocessed dataset used for analysis and visualization

3.3 SQL Script

- **File Name:** Capstone Customer Shopping.sql
- **Purpose:** Query-based analysis and insights generation

3.4 Power BI File

- **File Name:** Customer shopping dashboard.pbix
- **Purpose:** Interactive dashboard and reporting

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4. Data Dictionary

Table: Customer_Shopping

Column Name	Data Type	Description
invoice_no	Text	Unique invoice identifier
customer_id	Integer	Unique customer ID
gender	Text	Customer gender
age	Integer	Customer age
category	Text	Product category
quantity	Integer	Number of items purchased
price	Decimal	Price per item
payment_method	Text	Mode of payment
invoice_date	Date	Date of transaction
total_amount	Decimal	Calculated total bill amount

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5. Python Tasks Performed (customer_shopping.ipynb)

Data Cleaning

- Removal of null and duplicate records
- Data type correction
- Creation of derived columns (Total Amount)

```
[74]: df.isnull().sum()

[74]: Customer ID                0
      Age                        0
      Gender                     0
      Item Purchased             0
      Category                   0
      Purchase Amount (USD)      0
      Location                   0
      Size                       0
      Color                      0
      Season                     0
      Review Rating             37
      Subscription Status        0
      Shipping Type              0
      Discount Applied           0
      Promo Code Used            0
      Previous Purchases         0
      Payment Method             0
      Frequency of Purchases     0
      dtype: int64
```

```
df.isnull().sum()

Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating              0
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

Exploratory Data Analysis (EDA)

- **Data Loading**: Imported the dataset using pandas.

- **Initial Exploration**: Used df.info() to check structure and .describe() for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900.000000	3900	3900
2	NaN	6	7
No	NaN	PayPal	Every 3 Months
2223	NaN	677	584
NaN	25.351538	NaN	NaN
NaN	14.447125	NaN	NaN
NaN	1.000000	NaN	NaN
NaN	13.000000	NaN	NaN
NaN	25.000000	NaN	NaN
NaN	38.000000	NaN	NaN
NaN	50.000000	NaN	NaN

Output

- Clean dataset exported as `customer_shopping_clean.csv`

```
[90]: df
```

[90]:

	customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription_status
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes
...	...	...	...	...	...	...	...	...	...	...	...	...
3895	3896	40	Female	Hoodie	Clothing	28	Virginia	L	Turquoise	Summer	4.2	No
3896	3897	52	Female	Backpack	Accessories	49	Iowa	L	White	Spring	4.5	No
3897	3898	46	Female	Belt	Accessories	33	New Jersey	L	Green	Spring	2.9	No
3898	3899	44	Female	Shoes	Footwear	77	Minnesota	S	Brown	Summer	3.8	No
3899	3900	52	Female	Handbag	Accessories	81	California	M	Beige	Spring	3.1	No

3900 rows × 19 columns

```
[92]: # Export your manipulated DataFrame to CSV
df.to_csv("customer_shopping_clean.csv", index=False)
```

shipping_type	discount_applied	previous_purchases	payment_method	frequency_of_purchases	age_grp	purchase_frequency_days
Express	Yes	14	Venmo	Fortnightly	Middle_age	14
Express	Yes	2	Cash	Fortnightly	Young	14
Free Shipping	Yes	23	Credit Card	Weekly	Middle_age	7
Next Day Air	Yes	49	PayPal	Weekly	Young	7
Free Shipping	Yes	31	PayPal	Annually	Middle_age	365
...	...	...	...	...	...	...
2-Day Shipping	No	32	Venmo	Weekly	Adult	7
Store Pickup	No	41	Bank Transfer	Bi-Weekly	Middle_age	14
Standard	No	24	Venmo	Quarterly	Middle_age	90
Express	No	24	Venmo	Weekly	Adult	7
Store Pickup	No	33	Venmo	Quarterly	Middle_age	90

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6. SQL Tasks Performed

Data Analysis Queries

- Total revenue calculation by Gender

Result Grid | Filt

gender	Revenue
Male	157890
Female	75191

- Discounted Customers spend more the Avg amount

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69

- Top 5 highest avg review rating Products

item_purchased	Highest_avg_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

- Customer Segments count

customer_segments	Num_of_cust
Loyal	3116
Returning	701
New	83

- Revenue Contribution of each Age Group

age_grp	Revenue
Young	62143
Middle_age	59197
Adult	55978
Senior	55763

SQL Concepts Used

- SELECT, WHERE
- GROUP BY
- ORDER BY

- Aggregate functions (SUM, AVG, COUNT)
- CTE

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7. Power BI Data Model

Tables Used

- Customer_Shopping (cleaned dataset)

Relationships

- Single-table star model

Measures Created

- Total Revenue
- Average Order Value
- Total Quantity Sold
- Customer Count

Calculated Columns

- Age Group
- Revenue Category

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8. Dashboard Design (Images to be Added)

KPIs

- Total Revenue
- Total Customers
- Total Orders
- Average Order Value

Visualizations

- Category-wise revenue & sales bar chart
- Gender-wise revenue & sales distribution
- Percentage of Customers by Subscription
- Slicing by Subscription, Gender, Category & Shipping Type



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9. Business Insights

- Boost Subscriptions – Promote exclusive benefits for subscribers.
- Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.
- Review Discount Policy – Balance sales boosts with margin control.
- Product Positioning – Highlight top-rated and best-selling products in campaigns.
- Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.

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11. Conclusion

This capstone project demonstrates the effective use of SQL, Python, and Power BI to analyze customer shopping behavior. By integrating data cleaning, querying, and visualization, the project delivers valuable insights into customer preferences, revenue

drivers, and sales patterns. The analysis supports strategic decision-making in retail operations and provides a scalable foundation for future analytics enhancements.

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12. Author

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